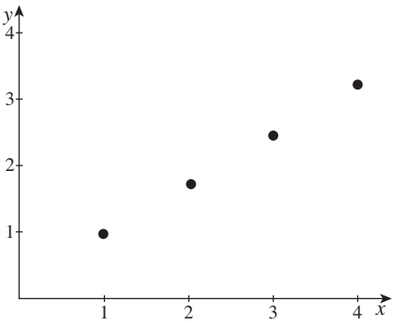


Figure 5.1

5.4.1 Arithmetic and Geometric Progressions

Progressions are some special cases of sequences where the terms of the sequences are either in increasing form or decreasing form.

We recall some definitions and results which has been discussed in earlier classes on arithmetic and geometric progressions.

Arithmetic Progression (AP)

- A sequence of the form

$$a,\ a+d,\ a+2d,\ a+3d,\ \ldots,\ a+(n-1)d,\ a+nd,\ \ldots$$

  is called an ***arithmetic progression*** or an ***arithmetic sequence***. In other words, each term (other than the first term) of the sequence is obtained by adding a constant to its previous term; the constant d is called ***common difference*** and the term a is called the ***initial term*** or ***first term***.
- The nth term of an arithmetic progression is given by $T_n = a + (n-1)d$.
- The sequences $\sqrt{2},\ \sqrt{2}+\sqrt{3},\ \sqrt{2}+2\sqrt{3},\ \sqrt{2}+3\sqrt{3},\ \ldots$ and $12,\ 9,\ 6,\ 3,\ \ldots$ are arithmetic sequences with common differences $\sqrt{3}$ and -3 respectively.
- It is interesting to observe that $3, 7, 11$ are three prime numbers which form an AP.
- For $n \in \mathbb{N}, T_n = an + b$ where a and b are relatively prime, form an AP which contains infinitely many prime numbers along with infinitely many composite numbers.

Geometric Progression (GP)

- A sequence of the form

$$a,\ ar,\ ar^2,\ ar^3,\ \ldots,\ ar^{n-1},\ ar^n,\ \ldots$$

  with $a \neq 0$, and $r \neq 0$ is called a ***geometric progression*** or a ***geometric sequence***. In other words, each term (other than the first term) of the sequence is obtained by multiplying its previous term by a constant; the constant r is called ***common ratio*** and the term a is called the ***initial term*** or ***first term***.
- The nth term of a geometric progression is given by $T_n = ar^{n-1}$.
- The sequences $1,\ 2,\ 4,\ 8,\ 16,\ \ldots$ and $\sqrt{2},\ 2,\ 2\sqrt{2},\ 4,\ 4\sqrt{2},\ 16,\ \ldots$ are geometric sequences with common ratios 2 and $\sqrt{2}$ respectively.
- Taking logarithm of each term in a geometric progression with positive common ratio yields an arithmetic progression. *i.e.*, If $a, ar, ar^2, \ldots$ is a GP with $r > 0$, then $\log a, \log(ar), \log(ar^2), \ldots$ is an AP with common difference $\log r$.

It is interesting to note that the constant sequence $c,\ c,\ c, \ldots$ is an arithmetic sequence and is also a geometric sequence if $c \neq 0$.

Let us consider the special constant sequence $0, 0, 0, 0, \ldots$. We have no problem in seeing this as an arithmetic sequence. But when we try to see this as a geometric sequence clearly the initial term